

Variational Autoencoder

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Data and Knowledge Modeling and Analysis (ECE 657A)
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Dataset Notations

training dataset: $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n, \mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$ (1)

embedding of training data (latent factors):

$$\{\mathbf{z}_i \in \mathbb{R}^p\}_{i=1}^n, \mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_n] \in \mathbb{R}^{p \times n} \quad (2)$$

$$p \leq d, \quad \text{usually: } p \ll d \quad (3)$$

Lecture Outline

- 1 Evidence Lower Bound
- 2 Expectation Maximization
- 3 Encoder of Variational Autoencoder
- 4 Decoder of Variational Autoencoder
- 5 Training VAE with Expectation Maximization
- 6 The Reparametrization Trick
- 7 Training VAE with Backpropagation
- 8 Examples

Evidence Lower Bound (ELBO)

$$\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i) = \frac{\mathbb{P}(\mathbf{x}_i | \mathbf{z}_i) \mathbb{P}(\mathbf{z}_i)}{\mathbb{P}(\mathbf{x}_i)} \quad (4)$$

$$\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}) = \frac{\mathbb{P}(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta}) \mathbb{P}(\mathbf{z}_i | \boldsymbol{\theta})}{\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta})} \quad (5)$$

$$\begin{aligned} \text{KL}(q(\mathbf{z}_i) \| \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta})) &= \int q(\mathbf{z}_i) \log \left(\frac{q(\mathbf{z}_i)}{\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta})} \right) d\mathbf{z}_i \\ &= \int q(\mathbf{z}_i) (\log(q(\mathbf{z}_i)) - \log(\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}))) d\mathbf{z}_i \\ &\stackrel{(5)}{=} \int q(\mathbf{z}_i) (\log(q(\mathbf{z}_i)) - \log(\mathbb{P}(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta})) \\ &\quad - \log(\mathbb{P}(\mathbf{z}_i | \boldsymbol{\theta})) + \log(\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta}))) d\mathbf{z}_i \end{aligned}$$

Evidence Lower Bound (ELBO)

$$\begin{aligned} &= \log(\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta})) + \int q(\mathbf{z}_i) (\log(q(\mathbf{z}_i)) \\ &\quad - \log(\mathbb{P}(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta})) - \log(\mathbb{P}(\mathbf{z}_i | \boldsymbol{\theta}))) d\mathbf{z}_i \\ &= \log(\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta})) + \int q(\mathbf{z}_i) \log\left(\frac{q(\mathbf{z}_i)}{\mathbb{P}(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta})\mathbb{P}(\mathbf{z}_i | \boldsymbol{\theta})}\right) d\mathbf{z}_i \\ &= \log(\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta})) + \int q(\mathbf{z}_i) \log\left(\frac{q(\mathbf{z}_i)}{\mathbb{P}(\mathbf{x}_i, \mathbf{z}_i | \boldsymbol{\theta})}\right) d\mathbf{z}_i \\ &= \log(\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta})) + \text{KL}(q(\mathbf{z}_i) \parallel \mathbb{P}(\mathbf{x}_i, \mathbf{z}_i | \boldsymbol{\theta})) \implies \end{aligned}$$

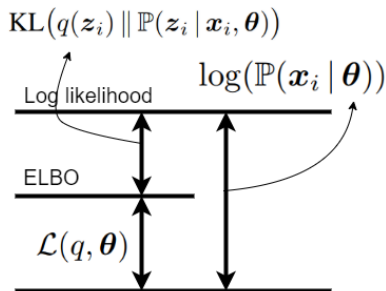
$$\log(\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta})) = \text{KL}(q(\mathbf{z}_i) \parallel \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta})) - \text{KL}(q(\mathbf{z}_i) \parallel \mathbb{P}(\mathbf{x}_i, \mathbf{z}_i | \boldsymbol{\theta}))$$

$$\mathcal{L}(q, \boldsymbol{\theta}) := -\text{KL}(q(\mathbf{z}_i) \parallel \mathbb{P}(\mathbf{x}_i, \mathbf{z}_i | \boldsymbol{\theta}))$$

$$\mathcal{L}(q, \boldsymbol{\theta}) = \log(\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta})) - \underbrace{\text{KL}(q(\mathbf{z}_i) \parallel \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}))}_{\geq 0} \leq \log(\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta}))$$

Evidence Lower Bound (ELBO)

$$\log(\mathbb{P}(\mathbf{x}_i | \boldsymbol{\theta})) = \text{KL}(q(\mathbf{z}_i) \parallel \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta})) + \mathcal{L}(q, \boldsymbol{\theta})$$



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Expectation Maximization Background

- *What shall we do? Some data are missing! The log-likelihood is not known completely so MLE cannot be used.*
- *Mmm, probably we can replace the missing data with something...*
- *Aha! Let us replace it with its mean.*
- *You are right! We can take the mean of log-likelihood over the possible values of the missing data. Then everything in the log-likelihood will be known, and then...*
- *And then we can do MLE!*

Expectation Maximization

$$\text{E-step: } q^{(t)} := \arg \max_q \mathcal{L}(q, \theta^{(t-1)}), \quad (6)$$

$$\text{M-step: } \theta^{(t)} := \arg \max_{\theta} \mathcal{L}(q^{(t)}, \theta), \quad (7)$$

E-step in EM for Variational Inference:

$$\begin{aligned} \max_q \mathcal{L}(q, \theta^{(t-1)}) &= \max_q \log(\mathbb{P}(\mathbf{x}_i | \theta^{(t-1)})) \\ &\quad + \max_q (-\text{KL}(q(\mathbf{z}_i) \| \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta^{(t-1)}))) \\ &= \max_q \log(\mathbb{P}(\mathbf{x}_i | \theta^{(t-1)})) + \min_q \text{KL}(q(\mathbf{z}_i) \| \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta^{(t-1)})) \end{aligned}$$

$$\text{KL}(q(\mathbf{z}_i) \| \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta^{(t-1)})) \stackrel{\text{set}}{=} 0 \implies \boxed{q(\mathbf{z}_i) = \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta^{(t-1)})}$$

Expectation Maximization

M-step in EM for Variational Inference:

$$\begin{aligned}\max_{\theta} \mathcal{L}(q^{(t)}, \theta) &= \max_{\theta} (-\text{KL}(q^{(t)}(z_i) \parallel \mathbb{P}(\mathbf{x}_i, z_i \mid \theta))) \\&= \max_{\theta} \left[- \int q^{(t)}(z_i) \log\left(\frac{q^{(t)}(z_i)}{\mathbb{P}(\mathbf{x}_i, z_i \mid \theta)}\right) dz_i \right] \\&= \max_{\theta} \int q^{(t)}(z_i) \log(\mathbb{P}(\mathbf{x}_i, z_i \mid \theta)) dz_i - \max_{\theta} \int q^{(t)}(z_i) \log(q^{(t)}(z_i)) dz_i.\end{aligned}$$

The second term is constant w.r.t. θ .

$$\begin{aligned}\max_{\theta} \mathcal{L}(q^{(t)}, \theta) &= \max_{\theta} \int q^{(t)}(z_i) \log(\mathbb{P}(\mathbf{x}_i, z_i \mid \theta)) dz_i \\&= \max_{\theta} \mathbb{E}_{\sim q^{(t)}(z_i)} [\log \mathbb{P}(\mathbf{x}_i, z_i \mid \theta)]\end{aligned}$$

$$\theta^{(t)} \leftarrow \arg \max_{\theta} \mathbb{E}_{\sim q^{(t)}(z_i)} [\log \mathbb{P}(\mathbf{x}_i, z_i \mid \theta)]$$

Expectation Maximization

EM for Variational Inference:

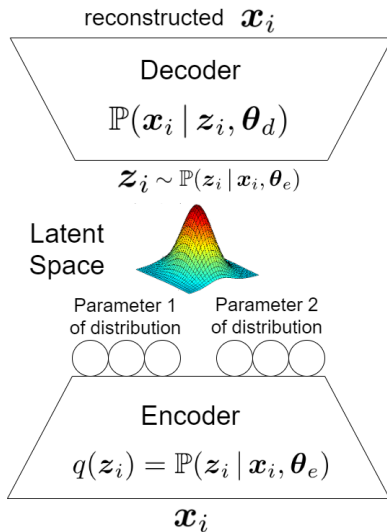
$$q^{(t)}(z_i) \leftarrow \mathbb{P}(z_i \mid \mathbf{x}_i, \boldsymbol{\theta}^{(t-1)}) \quad (8)$$

$$\boldsymbol{\theta}^{(t)} \leftarrow \arg \max_{\boldsymbol{\theta}} \mathbb{E}_{\sim q^{(t)}(z_i)} [\log \mathbb{P}(\mathbf{x}_i, z_i \mid \boldsymbol{\theta})] \quad (9)$$

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Parts of Encoder of Variational Autoencoder



Variational autoencoder was first proposed in [1].

Encoder of Variational Autoencoder

Encoder models:

$$q(\mathbf{z}_i) = \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}_e) \quad (10)$$

If this distribution is Gaussian:

$$q(\mathbf{z}_i) = \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}_e) = \mathcal{N}(\mathbf{z}_i | \boldsymbol{\mu}_{\mathbf{z}|\mathbf{x}}, \boldsymbol{\Sigma}_{\mathbf{z}|\mathbf{x}}) \quad (11)$$

In the case of multivariate Gaussian distribution for the latent space, the parameters are $\boldsymbol{\mu}_{\mathbf{z}|\mathbf{x}} = \mathbf{e}_1(\mathbf{x}_i, \boldsymbol{\theta}_{e,1})$ and $\boldsymbol{\Sigma}_{\mathbf{z}|\mathbf{x}} = \mathbf{diag}(\mathbf{e}_2(\mathbf{x}_i, \boldsymbol{\theta}_{e,2}))$.

Sampling from Latent Space:

$$\mathbf{z}_i \sim q(\mathbf{z}_i) = \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}_e). \quad (12)$$

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Decoder of Variational Autoencoder

Decoder models:

$$\mathbb{P}(\mathbf{x}_i \mid \mathbf{z}_i, \boldsymbol{\theta}_d) \quad (13)$$

to generate the data $\mathbf{x}_i$ from the latent factor $\mathbf{z}_i$.

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Training VAE with Expectation Maximization

$$\text{E-step: } \boldsymbol{\theta}_e^{(t)} := \arg \max_q \mathcal{L}(\boldsymbol{\theta}_e, \boldsymbol{\theta}_d^{(t-1)}, \mathbf{x}_i), \quad (14)$$

$$\text{M-step: } \boldsymbol{\theta}_d^{(t)} := \arg \max_q \mathcal{L}(\boldsymbol{\theta}_e^{(t)}, \boldsymbol{\theta}_d, \mathbf{x}_i). \quad (15)$$

$$\text{E-step: } \boldsymbol{\theta}_e^{(t)} := \boldsymbol{\theta}_e^{(t-1)} + \eta_e \frac{\partial \sum_{i=1}^b \mathcal{L}(\boldsymbol{\theta}_e, \boldsymbol{\theta}_d^{(t-1)}, \mathbf{x}_i)}{\partial \boldsymbol{\theta}_e}, \quad (16)$$

$$\text{M-step: } \boldsymbol{\theta}_d^{(t)} := \boldsymbol{\theta}_d^{(t-1)} + \eta_d \frac{\partial \sum_{i=1}^b \mathcal{L}(\boldsymbol{\theta}_e^{(t)}, \boldsymbol{\theta}_d, \mathbf{x}_i)}{\partial \boldsymbol{\theta}_d}, \quad (17)$$

Simplification of ELBO

The ELBO is simplified as:

$$\begin{aligned}\sum_{i=1}^b \mathcal{L}(q, \theta) &= - \sum_{i=1}^b \text{KL}(q(\mathbf{z}_i) \parallel \mathbb{P}(\mathbf{x}_i, \mathbf{z}_i \mid \theta_d)) \\ &\stackrel{(8)}{=} - \sum_{i=1}^b \text{KL}(\mathbb{P}(\mathbf{z}_i \mid \mathbf{x}_i, \theta_e) \parallel \mathbb{P}(\mathbf{x}_i, \mathbf{z}_i \mid \theta_d)) \\ &= - \sum_{i=1}^b \int \mathbb{P}(\mathbf{z}_i \mid \mathbf{x}_i, \theta_e) \log \left(\frac{\mathbb{P}(\mathbf{z}_i \mid \mathbf{x}_i, \theta_e)}{\mathbb{P}(\mathbf{x}_i, \mathbf{z}_i \mid \theta_d)} \right) d\mathbf{z}_i \\ &= - \sum_{i=1}^b \int \mathbb{P}(\mathbf{z}_i \mid \mathbf{x}_i, \theta_e) \log \left(\frac{\mathbb{P}(\mathbf{z}_i \mid \mathbf{x}_i, \theta_e)}{\mathbb{P}(\mathbf{x}_i \mid \mathbf{z}_i, \theta_d) \mathbb{P}(\mathbf{z}_i)} \right) d\mathbf{z}_i\end{aligned}$$

Simplification of ELBO

$$\begin{aligned}\sum_{i=1}^b \mathcal{L}(q, \theta) &= - \sum_{i=1}^b \int \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e) \log \left(\frac{\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e)}{\mathbb{P}(\mathbf{z}_i)} \right) d\mathbf{z}_i \\ &\quad + \sum_{i=1}^b \int \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e) \log (\mathbb{P}(\mathbf{x}_i | \mathbf{z}_i, \theta_d)) d\mathbf{z}_i \\ &= - \sum_{i=1}^b \text{KL}(\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e) \parallel \mathbb{P}(\mathbf{z}_i)) \\ &\quad + \sum_{i=1}^b \mathbb{E}_{\mathbf{z}_i \sim \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e)} \left[\log (\mathbb{P}(\mathbf{x}_i | \mathbf{z}_i, \theta_d)) \right].\end{aligned}\tag{18}$$

Simplification of ELBO

We found:

$$\begin{aligned}\sum_{i=1}^b \mathcal{L}(q, \theta) &= - \sum_{i=1}^b \text{KL}(\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e) \| \mathbb{P}(\mathbf{z}_i)) \\ &\quad + \sum_{i=1}^b \mathbb{E}_{\mathbf{z}_i \sim \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e)} \left[\log (\mathbb{P}(\mathbf{x}_i | \mathbf{z}_i, \theta_d)) \right].\end{aligned}$$

The second term in the above equation can be estimated using Monte Carlo approximation [2] where we draw ℓ samples $\{\mathbf{z}_{i,j}\}_{j=1}^{\ell}$ from $\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e)$:

$$\begin{aligned}\sum_{i=1}^b \mathcal{L}(q, \theta) &\approx \sum_{i=1}^b \tilde{\mathcal{L}}(q, \theta) = - \sum_{i=1}^b \text{KL}(\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e) \| \mathbb{P}(\mathbf{z}_i)) \\ &\quad + \sum_{i=1}^b \frac{1}{\ell} \sum_{j=1}^{\ell} \log (\mathbb{P}(\mathbf{x}_i | \mathbf{z}_{i,j}, \theta_d)).\end{aligned}\tag{19}$$

Simplification of ELBO

We found:

$$\begin{aligned}\sum_{i=1}^b \mathcal{L}(q, \theta) &\approx \sum_{i=1}^b \tilde{\mathcal{L}}(q, \theta) = - \sum_{i=1}^b \text{KL}(\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e) \| \mathbb{P}(\mathbf{z}_i)) \\ &\quad + \sum_{i=1}^b \frac{1}{\ell} \sum_{j=1}^{\ell} \log(\mathbb{P}(\mathbf{x}_i | \mathbf{z}_{i,j}, \theta_d)).\end{aligned}$$

$$\begin{aligned}\sum_{i=1}^b \mathcal{L}(q, \theta) &\approx \sum_{i=1}^b \tilde{\mathcal{L}}(q, \theta) = - \sum_{i=1}^b \mathbb{E}_{\sim \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e)} \left[\log \left(\frac{\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e)}{\mathbb{P}(\mathbf{z}_i)} \right) \right] \\ &\quad + \sum_{i=1}^b \frac{1}{\ell} \sum_{j=1}^{\ell} \log(\mathbb{P}(\mathbf{x}_i | \mathbf{z}_{i,j}, \theta_d)) \approx - \sum_{i=1}^b \frac{1}{\ell} \sum_{j=1}^{\ell} \log(\mathbb{P}(\mathbf{z}_{i,j} | \mathbf{x}_i, \theta_e)) \\ &\quad - \log(\mathbb{P}(\mathbf{z}_{i,j})) + \sum_{i=1}^b \frac{1}{\ell} \sum_{j=1}^{\ell} \log(\mathbb{P}(\mathbf{x}_i | \mathbf{z}_{i,j}, \theta_d)).\end{aligned}\tag{20}$$

Simplification of ELBO for Gaussian Distribution

We had in Eq. (19):

$$\begin{aligned}\sum_{i=1}^b \mathcal{L}(q, \theta) &\approx \sum_{i=1}^b \tilde{\mathcal{L}}(q, \theta) = - \sum_{i=1}^b \text{KL}(\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \theta_e) \| \mathbb{P}(\mathbf{z}_i)) \\ &\quad + \sum_{i=1}^b \frac{1}{\ell} \sum_{j=1}^{\ell} \log(\mathbb{P}(\mathbf{x}_i | \mathbf{z}_{i,j}, \theta_d)).\end{aligned}\quad (21)$$

For multivariate Gaussian distribution, we can simplify the KL divergence:

$$\begin{aligned}\sum_{i=1}^b \mathcal{L}(q, \theta) &\approx \sum_{i=1}^b \tilde{\mathcal{L}}(q, \theta) = - \sum_{i=1}^b \frac{1}{2} \left(\log\left(\frac{|\Sigma_z|}{|\Sigma_{z|x}|}\right) - \rho + \text{tr}(\Sigma_z^{-1} \Sigma_{z|x}) \right. \\ &\quad \left. + (\boldsymbol{\mu}_z - \boldsymbol{\mu}_{z|x})^\top \Sigma_z^{-1} (\boldsymbol{\mu}_z - \boldsymbol{\mu}_{z|x}) \right) + \sum_{i=1}^b \frac{1}{\ell} \sum_{j=1}^{\ell} \log(\mathbb{P}(\mathbf{x}_i | \mathbf{z}_{i,j}, \theta_d)).\end{aligned}\quad (22)$$

Training VAE with Expectation Maximization Using Approximations

$$\text{E-step: } \boldsymbol{\theta}_e^{(t)} := \boldsymbol{\theta}_e^{(t-1)} + \eta_e \frac{\partial \sum_{i=1}^b \tilde{\mathcal{L}}(\boldsymbol{\theta}_e, \boldsymbol{\theta}_d^{(t-1)}, \mathbf{x}_i)}{\partial \boldsymbol{\theta}_e}, \quad (23)$$

$$\text{M-step: } \boldsymbol{\theta}_d^{(t)} := \boldsymbol{\theta}_d^{(t-1)} + \eta_d \frac{\partial \sum_{i=1}^b \tilde{\mathcal{L}}(\boldsymbol{\theta}_e^{(t)}, \boldsymbol{\theta}_d, \mathbf{x}_i)}{\partial \boldsymbol{\theta}_d}, \quad (24)$$

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The Reparametrization Trick

Sampling the ℓ samples for the latent variables, i.e. Eq. (12), blocks the gradient flow because computing the derivatives through $\mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}_e)$ by chain rule gives a high variance estimate of gradient.

To overcome this problem, we use the reparameterization technique [1, 3]. In this technique, instead of sampling $\mathbf{z}_i \sim \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}_e)$, we assume $\mathbf{z}_i$ is a random variable but is a deterministic function of another random variable ϵ_i as follows:

$$\mathbf{z}_i = g(\epsilon_i, \mathbf{x}_i, \boldsymbol{\theta}_e) \quad (25)$$

$$\epsilon_i \sim \mathbb{P}(\epsilon) \quad (26)$$

$$\mathbb{E}_{\sim \mathbb{P}(\mathbf{z}_i | \mathbf{x}_i, \boldsymbol{\theta}_e)}[f(\mathbf{z}_i)] \rightarrow \mathbb{E}_{\sim \mathbb{P}(\epsilon_i | \mathbf{x}_i, \boldsymbol{\theta}_e)}[f(g(\epsilon_i, \mathbf{x}_i, \boldsymbol{\theta}_e))] \quad (27)$$

An example:

$$z_i \sim \mathcal{N}(\mu, \sigma^2), \quad \epsilon_i \sim \mathcal{N}(0, 1), \quad z_i = g(\epsilon_i) = \mu + \sigma \epsilon_i.$$

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Training VAE with Backpropagation

In practice, VAE is trained by backpropagation [3] where the backpropagation algorithm is used.

If $\theta := \{\theta_e, \theta_d\}$, we have:

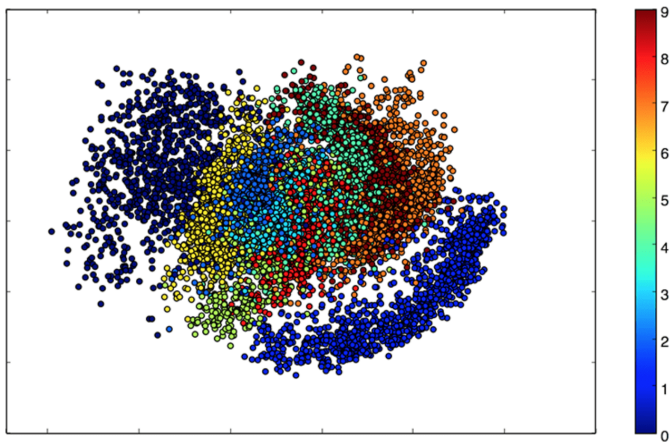
$$\theta^{(t)} := \theta^{(t-1)} - \eta \frac{\partial \sum_{i=1}^b -\tilde{\mathcal{L}}(\theta, \mathbf{x}_i)}{\partial \theta} \quad (28)$$

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Example on MNIST Dataset

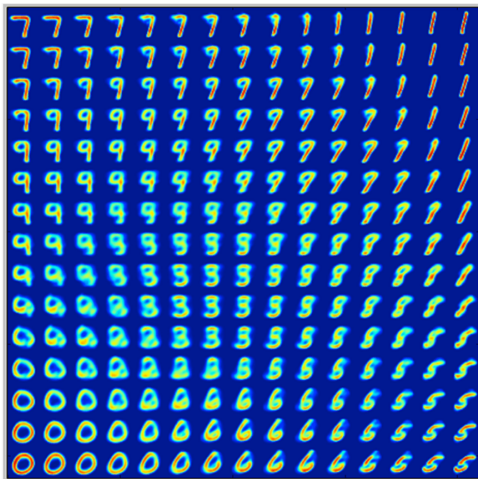
The embedding of MNIST dataset in the latent space:



The credit of this image is for: <https://blog.keras.io/building-autoencoders-in-keras.html>

Example on MNIST Dataset

The 2D manifold of variational autoencoder:



The credit of this image is for: <https://blog.keras.io/building-autoencoders-in-keras.html>

Useful Resources To Read

- Tutorial paper: “Factor Analysis, Probabilistic Principal Component Analysis, Variational Inference, and Variational Autoencoder: Tutorial and Survey” [4]
- Tutorial YouTube videos by Carnegie Mellon University, Deep Learning: [\[Click here\]](#)
- Tutorial paper: “Tutorial on variational autoencoders” [5]
- Keras code for VAE: [\[Click here\]](#)

References

- [1] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *International Conference on Learning Representations*, 2014.
- [2] B. Ghojogh, H. Nekoei, A. Ghojogh, F. Karray, and M. Crowley, "Sampling algorithms, from survey sampling to Monte Carlo methods: Tutorial and literature review," *arXiv preprint arXiv:2011.00901*, 2020.
- [3] D. J. Rezende, S. Mohamed, and D. Wierstra, "Stochastic backpropagation and approximate inference in deep generative models," in *International Conference on Machine Learning*, 2014.
- [4] B. Ghojogh, A. Ghodsi, F. Karray, and M. Crowley, "Factor analysis, probabilistic principal component analysis, variational inference, and variational autoencoder: Tutorial and survey," *arXiv preprint arXiv:2101.00734*, 2021.
- [5] C. Doersch, "Tutorial on variational autoencoders," *arXiv preprint arXiv:1606.05908*, 2016.